

Home Appliances Repair & Service Training

2 Days Professional Training Program (Theory + Practical)

2-Day Training Overview

1

Day 1 – Microwave Oven & Water Heater

Deep-dive into microwave oven components including **magnetron, high voltage capacitor, transformer**, and door switch testing. Plus complete water heater (geyser) service covering heating elements, thermostats, and practical repair.

2

Day 2 – RO Purifier, Kitchen Chimney & Air Cooler

Hands-on training with **RO water purifier systems** including booster pump and SMPS, followed by kitchen chimney motor and filter servicing, and complete air cooler maintenance and motor testing.

3

Theory + Practical Approach

Every session combines structured theory on working principles and components with **live practical troubleshooting** so trainees gain real-world fault-finding and repair confidence before they leave the classroom.

Microwave Oven: Theory & Practical

Introduction & Working Principle

Overview of microwave oven **types and classification**. Understanding the working principle of microwave energy generation and cooking. Identification of all major internal and external components before any repair attempt.

Key Components In-Depth

Detailed study of the **Magnetron** (the core microwave generator), **High Voltage Capacitor**, and **High Voltage Transformer**. Understanding their roles, failure modes, safe discharge procedures, and testing methodologies.

Testing, Fault Finding & Safety

Door Switch Testing and **Fuse Testing** with multimeter. Common fault diagnosis — no heat, sparking, turntable issues, display failure. Critical **safety precautions** for high-voltage circuits followed by live practical troubleshooting sessions.

Water Heater (Geyser): Theory & Practical



Types & Core Components

Introduction to **Storage** and **Instant Water Heaters** — design differences and application scenarios. Study of the Heating Element, Thermostat, and Thermal Cut-out: function, specifications, and failure identification.



Electrical Wiring & Component Testing

Understanding the **complete electrical wiring diagram** of storage and instant geysers. Practical testing of **Heating Element** with continuity test, **Thermostat** temperature calibration check, and thermal cut-out reset and replacement.



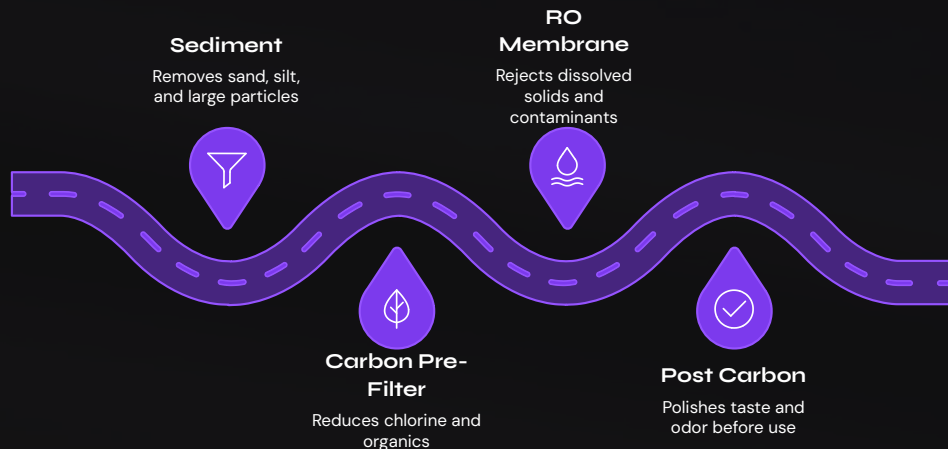
Water Leakage & Practical Repair

Identifying sources of **water leakage** — inlet/outlet fittings, pressure relief valve, tank body, and element seals. Systematic step-by-step practical repair and service including **safe draining, element replacement, and pressure testing**.

RO Water Purifier: Theory & Practical

System Overview & Working Principle

Introduction to the **Reverse Osmosis system** — how water is forced through semi-permeable membranes to remove dissolved solids, bacteria, and contaminants. Complete **filter identification**: sediment, carbon pre-filter, RO membrane, and post-carbon.



Key Components & Servicing

- **Booster Pump** – testing voltage, pressure output, and motor winding
- **Solenoid Valve** – function, testing for stuck-open/closed faults
- **SMPS** – switched-mode power supply output voltage testing
- **TDS Controller** – adjusting mineral retention levels correctly
- **Water Leakage Repair** – identifying fitting and membrane housing leaks
- **Filter Replacement** – replacement schedule, cartridge fitting, system flush
- **Practical Service** – complete service strip-down and reassembly

Kitchen Chimney: Theory & Practical



Types & Classification

Wall-mounted, island, built-in, and ductless chimneys. Differences in airflow design, motor capacity, and filter types for each chimney category.



Motor, Blower & Capacitor

Understanding the **induction motor** driving the blower, the role of the **run capacitor** in starting torque, and blower blade inspection for grease buildup and imbalance faults.



Switch Panel & Testing

Diagnosis of **switch panel** faults — speed selector, light switch, and touch-control boards. **Motor testing** with multimeter for winding resistance and capacitor ESR check.



Oil & Filter Cleaning

Safe removal and **deep cleaning of oil collector trays and baffle/mesh filters**. Practical service walkthrough — disassembly, degreasing, reassembly, and performance verification after service.

Air Cooler: Theory & Practical

Working Principle & Components

How evaporative cooling works — hot air drawn through **water-saturated cooling pads** drops temperature via evaporation. Key components covered:

1 Fan Motor & Speed Control Switch

Motor winding testing, three-speed selector switch fault diagnosis, and capacitor role in motor starting.

2 Water Pump & Distribution System

Submersible pump testing for flow rate and voltage draw. Water distribution pipes, spider fitting inspection, and pad saturation verification.

3 Cooling Pad Inspection

Identifying clogged, hardened, or algae-damaged honeycomb pads. Cleaning vs. replacement decision criteria.

Testing & Complete Service Practice

Step-by-step practical service covering the full air cooler:

- **Motor Testing** – continuity, winding resistance, run capacitor check
- **Pump Testing** – current draw, impeller blockage, output flow test
- **Speed Control Switch** – contact continuity at each speed position
- **Capacitor Testing** – capacitance measurement and visual inspection
- **Water Level Float** – auto-fill valve function and adjustment
- **Complete Service** – full strip-down, cleaning, lubrication, and reassembly with functional test run

PROGRAM SUMMARY

What You Will Achieve

5 Appliances

Complete repair and service skills across Microwave Oven, Water Heater, RO Purifier, Kitchen Chimney, and Air Cooler.

Theory + Hands-On

Every module pairs component-level theory with live practical fault-finding and service sessions.

Job-Ready Skills

Walk away with the confidence and competence to diagnose faults, test components, and carry out professional repairs.